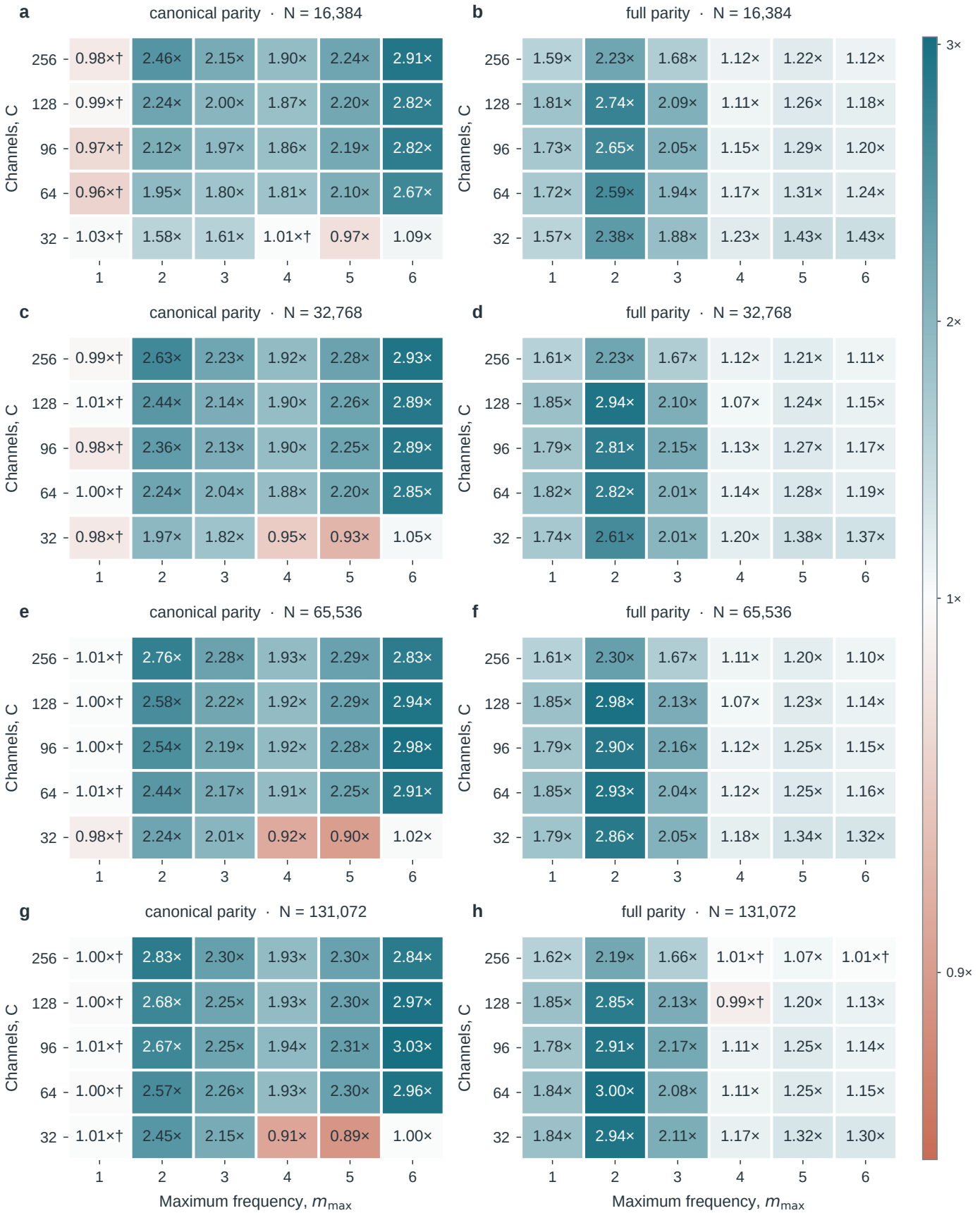
ICTC-O2 HP vs eqx: throughput

O(2) tensor products · Forward · N = 16,384–131,072 · C = 32–256 · m_max = 1–6



a	1	2	3	4	5	6
256	0.98×†	2.46×	2.15×	1.90×	2.24×	2.91×
128	0.99×†	2.24×	2.00×	1.87×	2.20×	2.82×
96	0.97×†	2.12×	1.97×	1.86×	2.19×	2.82×
64	0.96×†	1.95×	1.80×	1.81×	2.10×	2.67×
32	1.03×†	1.58×	1.61×	1.01×†	0.97×	1.09×

b	1	2	3	4	5	6
256	1.59×	2.23×	1.68×	1.12×	1.22×	1.12×
128	1.81×	2.74×	2.09×	1.11×	1.26×	1.18×
96	1.73×	2.65×	2.05×	1.15×	1.29×	1.20×
64	1.72×	2.59×	1.94×	1.17×	1.31×	1.24×
32	1.57×	2.38×	1.88×	1.23×	1.43×	1.43×

c	1	2	3	4	5	6
256	0.99×†	2.63×	2.23×	1.92×	2.28×	2.93×
128	1.01×†	2.44×	2.14×	1.90×	2.26×	2.89×
96	0.98×†	2.36×	2.13×	1.90×	2.25×	2.89×
64	1.00×†	2.24×	2.04×	1.88×	2.20×	2.85×
32	0.98×†	1.97×	1.82×	0.95×	0.93×	1.05×

d	1	2	3	4	5	6
256	1.61×	2.23×	1.67×	1.12×	1.21×	1.11×
128	1.85×	2.94×	2.10×	1.07×	1.24×	1.15×
96	1.79×	2.81×	2.15×	1.13×	1.27×	1.17×
64	1.82×	2.82×	2.01×	1.14×	1.28×	1.19×
32	1.74×	2.61×	2.01×	1.20×	1.38×	1.37×

e	1	2	3	4	5	6
256	1.01×†	2.76×	2.28×	1.93×	2.29×	2.83×
128	1.00×†	2.58×	2.22×	1.92×	2.29×	2.94×
96	1.00×†	2.54×	2.19×	1.92×	2.28×	2.98×
64	1.01×†	2.44×	2.17×	1.91×	2.25×	2.91×
32	0.98×†	2.24×	2.01×	0.92×	0.90×	1.02×

f	1	2	3	4	5	6
256	1.61×	2.30×	1.67×	1.11×	1.20×	1.10×
128	1.85×	2.98×	2.13×	1.07×	1.23×	1.14×
96	1.79×	2.90×	2.16×	1.12×	1.25×	1.15×
64	1.85×	2.93×	2.04×	1.12×	1.25×	1.16×
32	1.79×	2.86×	2.05×	1.18×	1.34×	1.32×

g	1	2	3	4	5	6
256	1.00×†	2.83×	2.30×	1.93×	2.30×	2.84×
128	1.00×†	2.68×	2.25×	1.93×	2.30×	2.97×
96	1.01×†	2.67×	2.25×	1.94×	2.31×	3.03×
64	1.00×†	2.57×	2.26×	1.93×	2.30×	2.96×
32	1.01×†	2.45×	2.15×	0.91×	0.89×	1.00×

h	1	2	3	4	5	6
256	1.62×	2.19×	1.66×	1.01×†	1.07×	1.01×†
128	1.85×	2.85×	2.13×	0.99×†	1.20×	1.13×
96	1.78×	2.91×	2.17×	1.11×	1.25×	1.14×
64	1.84×	3.00×	2.08×	1.11×	1.25×	1.15×
32	1.84×	2.94×	2.11×	1.17×	1.32×	1.30×

† IQR overlap * isolated timing ‡ memory increase